05-05 第2篇

阅读理解（说明文） How Does Soap Really Clean Our Hands?

① Every morning before breakfast, a teacher or parent says the same simple words: "Wash your hands with soap." Most students nod and do it. But have you ever stopped and asked, "Why doesn't just water work? Why do we need soap?" The answer is a small but clever story about chemistry and the nature of dirt.

② The problem starts with **oil**. Our hands are covered with tiny amounts of oil all the time, from our skin and from the things we touch — schoolbags, door handles, fried food, phone screens. Dust, germs, and small bits of food like to stick to this oil. Pure water and oil do not mix. If you only rinse your hands with water, the oil stays, and most of the germs stay with it.

③ This is where soap comes in. A single **soap molecule** is a very special little tool. It has two ends: one end is happy to meet water, and the other end is happy to meet oil. Scientists call them the "water-loving head" and the "oil-loving tail." When soap meets water on your skin, millions of these little molecules form small round balls, called **micelles**.

④ During **rubbing**, the oil-loving tails of the soap push into the oil layer on your hands and break it into tiny drops. Each tiny oil drop is then trapped inside a ball of soap molecules, with the water-loving heads pointing outward. Now the oil, along with the dust and germs stuck to it, is wrapped up and ready to leave.

⑤ This is why the next step is so important: **rinsing**. Clean running water washes the soap balls off your skin and down the drain. Germs travel with them. If you do not rinse well, some of the trapped dirt stays and may even move to your towel. That is why doctors say: wet hands, soap well, rub for 20 seconds, and rinse fully.

⑥ Hot water is not strictly needed. Cold water works, as long as you use enough soap and rub long enough. However, soap cannot remove every kind of virus. That is why washing hands is often paired with other care, like wearing clean clothes and avoiding coughing into open hands. Next time you see a bar of soap by the sink, remember: it is not magic. It is a very polite little "taxi" that carries dirt safely away from your skin.

根据短文内容，选择最佳答案。

1. Why doesn't pure water clean hands very well?

A. Water is too cold.
B. Water and oil do not mix, so the oil and germs on skin stay.
C. Water is too thick.
D. Water cannot touch skin.

2. What do the two ends of a soap molecule like?

A. Both ends like oil.
B. Both ends like water.
C. One end likes water, the other end likes oil.
D. Neither end likes anything.

3. What happens during rubbing with soap?

A. The oil turns into sugar.
B. The oil-loving tails push into the oil layer and break it into tiny drops.
C. The water freezes.
D. The soap disappears into air.

4. What does the writer compare soap to in the last paragraph?

A. A magic wand.
B. A sharp knife.
C. A polite little "taxi".
D. A loud drum.

5. What is the passage MAINLY about?

A. How to grow sunflowers at home.
B. Why chemistry is the hardest subject at school.
C. How soap cleans hands with the help of special molecules.
D. Why students should study harder in spring.

答案与解析

1.
【答案】B
【题型】细节理解题
【定位】第②段末句 "Pure water and oil do not mix... most of the germs stay with it."
【解析】水油不相溶，因此只用水无法带走油中细菌。A 水冷、C 水厚、D 水不能触皮均与原文不符。故选 B。

2.
【答案】C
【题型】细节理解题

【定位】第③段第3-4句 "water-loving head" 与 "oil-loving tail"
【解析】肥皂分子一端亲水、一端亲油。A/B/D 与原文不符。故选 C。

3.
【答案】B
【题型】细节理解题
【定位】第④段第1句 "the oil-loving tails of the soap push into the oil layer... break it into tiny drops."
【解析】亲油端钻入油层并将油打碎成小滴。A 变糖、C 水结冰、D 肥皂消失均无依据。故选 B。

4.
【答案】C
【题型】细节理解/写作意图题
【定位】第⑥段末句 "a very polite little 'taxi' that carries dirt safely away"
【解析】作者将肥皂比作把脏物带走的"出租车"。A 魔杖、B 刀、D 鼓均不符原文比喻。故选 C。

5.
【答案】C
【题型】主旨大意题
【定位】全文各段围绕"肥皂为何能洗净手"展开
【解析】文章讲述肥皂分子如何借亲水亲油特性带走污垢与细菌。A 种葵花、B 化学难学、D 努力学习都与中心不符。故选 C。

词汇注释（超纲词）

- **molecule** [n.] 分子
- **micelle** [n.] 胶束（肥皂分子形成的小球）
- **virus** [n.] 病毒
- **rinse** [v.] 冲洗